

SHRI G.S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
SESSION: JULY 2025- DEC 2025; Semester "A"
IT 10007 Fundamentals of IT & AI
MST-II

Max Marks:20

Time: 60 min

Q. No.	Question	MARKS	BL	CO
1	Explain what are the different data types in 'C' programming language.	2	1	2
2	Fill in the blanks in the following: i. == is a _____ operator ii. _____ case is matched in a switch statement, when all cases are unmatched.	2	1	2
3	Write a C program to enter a number and then calculate the sum of its digits. Sample Input: 123 Sample output: 6	4	3	2
4	Compare the following i. Call by value and call by reference parameter passing in a function. ii. Scope of local and global variables in a function. iii. Run-time error and compile-time error.	6	2	3
5	Consider a positive integer n. The following code determines whether n can be expressed as a power of 4. Fill up the dashed lines in the program. <pre>#include<stdio.h> int main (){ int n, temp, flag=0; scanf("%d",&n); temp=n; while(____) { if(____) { flag=1; break;} temp=____;} if(____) printf("%d is a number that can be expressed as power of 4.",n); else printf("%d can not be expressed as power of 4",n); }</pre>	2	3	3
6	Write the if-else-if equivalent code for the following code snippet: <pre>#include<stdio.h> int main(){ int number, x=5; scanf("%d", &number); switch(number) { case2:{printf("case two"); x=x*12;} case21:{printf("case twenty one");x=x+2;break;} case 22: { printf("case twenty two"); x = x-5;} case1:{printf("case one");continue;} default :{ printf("Nothing"); } } printf("%d",x); return 0; }</pre>	4	3	3